Distributed Systems Scenario-Based Questions - Practice Questions Justify your choice

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Question 1 – Collaborative Code Editor (e.g., Google Docs, but for Legal Contracts)

A team of lawyers simultaneously edits a high-stakes legal contract in a real-time collaborative editor. Every keystroke from any user must be seen by all others in the exact same order. If two lawyers edit the same paragraph at the same time, the system must merge changes deterministically without conflicts. The system is distributed across three data centers. Network splits are rare but possible. The company is willing to sacrifice a few seconds of responsiveness to avoid corrupted legal text.

Question: Which model should the backend document storage prioritize?

- A) CAP – Consistency (CP)
- B) CAP – Availability (AP)
- C) ACID (on a single leader with synchronous replication)
- D) BASE with strong eventual consistency

Question 2 – Smart Factory Defect Detector

A factory produces millions of microchips per hour. Each chip passes under a high-speed camera that logs its serial number and a "defect probability" score. The system must write every single score to a distributed database for offline quality analysis. If one write fails (e.g., network timeout), the system must still accept the next chip's data without waiting. Losing a few scores (0.01%) is acceptable because the factory already uses statistical sampling. However, the system cannot stop or slow down the production line.

Question: Which model best fits the write path of this system?

- A) ACID transactions
- B) CAP – Consistency (CP)
- C) CAP – Availability (AP)
- D) BASE with no cross-node coordination

Question 3 – Air Traffic Control Secondary Radar (Mode S)

An air traffic control system receives aircraft position reports (Mode S transponder messages) every second. The system must maintain a global view of all aircraft within a 200-mile radius. If two radar stations report conflicting positions for the same aircraft due to a network partition, the system must discard both reports and flag an alert rather than showing a possibly wrong location. Controllers trust the system only when all data sources agree.

Question:
Which model should the data fusion engine follow?

- A) CAP – Availability (AP)
- B) CAP – Consistency (CP)
- C) ACID (with two-phase commit)
- D) BASE (last write wins)

Question 4 – Subway Fare Gate System (Offline Mode)

A subway system uses electronic fare gates that must work even when the central server is unreachable (e.g., tunnel network outage). Each gate stores a local, encrypted list of valid passes. When a commuter taps a card, the gate must approve or deny entry immediately – waiting for the server is not allowed. The gate may occasionally approve a pass that was recently blocked (a few seconds delay until the gate syncs), but the risk of lost revenue is low. The alternative – denying entry to valid passes – would cause passenger chaos.

Question: Which model should the fare gate's local validation logic use?

- A) ACID (distributed transaction to central DB)
- B) CAP – Consistency (CP)
- C) CAP – Availability (AP)
- D) BASE with optimistic update and periodic sync

Question 5 – Scientific Observatory Data Archive (Raw Neutron Capture Events)

A neutron telescope at a polar research station records 1,000 events per second. Each event is timestamped and written to a distributed storage system across two mirrored nodes. The experiment requires that no event is ever lost – even a single missing neutron capture invalidates statistical analysis. The network connection is unreliable (ice storms). However, the system can pause recording for minutes if needed, as long as no data is permanently lost once the connection recovers.

Question:
Which model should the write acknowledgement logic follow?

- A) CAP – Availability (AP)
- B) BASE (asynchronous write)
- C) CAP – Consistency (CP) with quorum write (2 out of 2 nodes)
- D) ACID on a single node only (no distribution)

Question 6 – Online Exam Proctoring Suspicion Log

Scenario:
An online exam platform logs every suspicious behaviour (tab switch, background process, face mismatch) into a distributed "cheating suspicion" table. This log is used only for post-exam review by human proctors; it is never accessed during the exam. The system handles 100,000 concurrent exams. Network partitions are common, but losing a log entry is acceptable (proctors may miss one clue among many). The absolute priority is that the exam itself does not freeze or crash for students.
Question:
Which model should the suspicion logging service adopt?

- A) ACID (synchronous commit)
- B) CAP – Consistency (CP)
- C) CAP – Availability (AP)
- D) BASE (fire-and-forget, eventual persistence)

Question 7 – Drone Swarm Formation Control (Military Reconnaissance)

A swarm of 50 drones flies over enemy territory. Each drone continuously shares its position and target lock status. All drones must agree on a single formation command (e.g., "turn left 30°") within 200 ms, otherwise the swarm could collide or miss the target. If some drones cannot communicate due to jamming, the rest must still execute a consistent decision – they cannot each decide independently. The system can tolerate short loss of control inputs, but inconsistent commands are unacceptable.
Question:
Which model should the distributed consensus mechanism use?

- A) CAP – Availability (AP) – let each drone decide
- B) BASE (optimistic update)
- C) CAP – Consistency (CP) with majority quorum
- D) ACID (two-phase commit)

Question 8 – Mobile Banking Fund Transfer

A mobile banking application allows customers to transfer money between accounts in real time. When a user transfers $5,000 from Account A to Account B, the system must ensure that the money is deducted from one account and added to the other exactly once. If a network failure occurs during the transfer, the transaction must not leave the system in a partially updated state. Customers may tolerate a short delay during peak hours, but incorrect balances or duplicate transfers are unacceptable.

Question:
Which model should the transaction processing system prioritize?

- A) CAP – Availability (AP)
- B) BASE with eventual consistency
- C) ACID transactions with strong consistency
- D) CAP – Availability (AP) with asynchronous replication

Question 9 – Social Media News Feed

A global social media platform allows users to post photos, comments, and status updates. Millions of users continuously refresh their feeds. During temporary network failures, users should still be able to view and upload posts even if some friends' updates appear a few seconds late. The platform prioritizes responsiveness and uninterrupted user experience over perfectly synchronized timelines.

Question:
Which model best fits the feed distribution system?

- A) ACID transactions with synchronous replication
- B) CAP – Consistency (CP)
- C) CAP – Availability (AP)
- D) BASE with eventual consistency